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Evaluation of IoT based smart safety systems for women and children using machine learning techniques

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Traditional security systems for women and children often fail, limited by manual activation and slow response times. This research presents an IoT-enabled smart safety framework that leverages machine learning (ML) to autonomously detect and respond to physiological distress and potential threats. The proposed system integrates physiological sensors (heart rate, temperature) and activity sensors (GPS, accelerometer) into an intelligent wearable device. A hybrid ML approach, primarily utilizing Support Vector Machine (SVM) and Naive Bayes (NB), is employed for robust activity recognition and stress level classification. Performance was rigorously validated using k-fold cross-validation, with the SVM classifier achieving a 99.7% average accuracy in threat detection. This AI-driven approach reduces detection latency to 3 s, while battery optimization ensures 18–20 h of continuous operation. Upon autonomous threat detection, the system uses GSM connectivity to transmit GPS coordinates to authorities. This research demonstrates a practical, high-accuracy solution for personal security, with a strong emphasis on data privacy and ethical deployment.

Keywords Body sensors, Arduino, Raspberry pi, GPS, Internet of things, Women security, Smart band

The escalating incidence of crimes targeting women and children has underscored the urgent need for innovative technological interventions capable of delivering real-time protection and rapid emergency response[cite: 84]. Conventional safety mechanisms, which often depend on manual activation or post-incident reporting, fall short in preventing harm or ensuring timely assistance during critical moments [cite: 85]. Many current solutions require explicit user activation—an impractical expectation during emergencies when victims may be incapacitated or unable to trigger an alert [cite: 89]. While the integration of Internet of Things (IoT) technologies offers transformative opportunities [cite: 86], existing applications face significant challenges, including delayed response times, a lack of continuous monitoring, and poor integration with emergency services [cite: 88].

A more fundamental gap, however, lies in the **logic** of threat detection. Many systems oversimplify the problem, assuming that a single physiological anomaly, such as an elevated heart rate, directly correlates with an unsafe situation. This fails to account for real-world variability—for example, the body's natural response to exercise or daily stress—and leads to a high rate of false positives, which erodes user trust and system reliability. This research addresses this critical gap by proposing a robust framework that integrates IoT, cloud computing, and advanced machine learning (ML) to create an autonomous, context-aware safety system.

The system moves beyond simple threshold-based alerts by using a hybrid ML approach (combining SVM, Naive Bayes, and CNN classifiers) to analyze a fusion of sensor data [cite: 82]. By monitoring multiple inputs—including heartbeat monitors, temperature sensors, accelerometers, and GPS modules—the system learns personalized physiological baselines to accurately distinguish between benign stress and genuine distress scenarios [cite: 72, 74]. Recent studies have validated the potential for multimodal sensor analysis in security, confirming that this is a promising research direction. [cite: 18].

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Literature survey

Existing methodologies

Numerous researchers have explored the application of Information and Communication Technology (ICT) to support and protect survivors of Intimate Partner Violence (IPV)[1]. IPV is a widespread global public health crisis and a serious human rights violation, characterized by patterns of controlling, coercive, threatening, or violent behavior perpetrated by current or former intimate partners. Studies show that IPV affects approximately 30% of women globally over their lifetime, with profound consequences for both physical and mental health.

The complexity of IPV includes physical assault, psychological abuse, sexual coercion, financial manipulation, and increasingly, digital harassment. These forms of abuse often result in long-term trauma that persists beyond the immediate threat. Research has identified several contributing factors to IPV, such as childhood exposure to violence, substance abuse, financial stress, and cultural norms that reinforce gender inequality. The COVID-19 pandemic further intensified the frequency and severity of IPV incidents, emphasizing the need for comprehensive prevention strategies, improved screening in healthcare settings, and trauma-informed interventions that address both immediate safety and long-term recovery.

Technological solutions proposed in recent studies include cloud computing, home automation, artificial intelligence (AI), machine learning (ML), and other innovative approaches. This research focuses on the potential of intelligent biosensors to autonomously detect violence and trigger alerts, offering a proactive safety mechanism. Previous attempts in this domain provide valuable insights.

Hany et al.[2] suggested monitoring electrodermal activity, electroencephalogram (EEG), and electrocardiogram (ECG) signals to detect hostility. Patel and Hassan[3] proposed using an altimeter embedded in wearable devices for aggression detection, although their approach was limited to single-sensor input. Goodwin introduced a multi-sensor method combining cardiovascular, existential, and motion (accelerometry) data, specifically targeting individuals with Down syndrome. However, this model lacks generalizability to IPV scenarios.

While some proposals offer comprehensive frameworks[4], many rely on limited features and lack continuous surveillance capabilities. Kumari's work, for instance, focuses on tracking daily activities that may include offensive behavior, but does not specifically address the needs of IPV survivors.

Self-management in IPV contexts involves continuous data collection to monitor survivor progress and detect potential threats. Advances in biometrics and IoT technologies enable 24/7 monitoring through mobile devices, capturing critical physiological data. Wu et al.[5] demonstrated how activity monitoring systems can assess independence and functional capacity, serving as indicators of well-being.

Body Area Networks (BANs) represent another promising solution. These wearable systems, positioned near the body, track biophysical signals to detect aggressive behavior. Broens et al.[6] first introduced BANs during asthma research, proposing 24-hour monitoring to anticipate attacks. Later enhancements incorporated data management for improved analysis[7]. BANs typically include accelerometers, GPS modules, and communication interfaces—features now commonly found in smartphones.

Emotion analysis also plays a key role, utilizing respiration patterns and distress signals (e.g., crying, laughing) to establish physiological trends. Parameters such as volume, intensity, and pacing help identify emotional states across a spectrum from calm to stressed[8]. Breath-based sensors have shown promise in non-invasive health monitoring and abuse detection, although their adoption remains limited despite two decades of research[9].

Most wearable technologies now include heart rate (HR) monitoring. Smart bands often use photoplethysmography, a technique that shines light on the skin to measure blood flow, contrasting with ECG, which tracks electrical signals. Devices like Spotify-integrated fitness trackers offer precision suitable for IPV monitoring[10]. The medical community has long recognized the link between chronic stress and elevated heart rate, reinforcing the relevance of HR monitoring in IPV detection[11].

Recent studies suggest that fluctuations in glycaemia may serve as diagnostic indicators for the presence or risk of violence in individuals. Research has established correlations between glucose levels, stress-induced hormonal imbalances, and blood variables associated with hyperglycemia. Additionally, a notable relationship exists between obesity and elevated levels of blood oxaloacetate, which tend to spike during periods of acute stress[12].

Skin Conductance (SC) testing, which measures the skin's electrical activity through applied sensors, has proven useful in assessing physiological responses. SC readings are commonly used in polygraph examinations to evaluate emotional arousal, with lower SC values indicating reduced excitement levels[13]. A correlation between intoxication and SC has also been observed, as thermal conductivity changes reflect central nervous system activity. Affordable SC-based sensing devices such as the Empatica E4, Microsoft Band 2, and the eHealth Sensor Platform with My Signals have demonstrated reliable performance. Notably, both the Empatica E4 and Microsoft Band 2 support Ethernet connectivity for enhanced data transmission[14].

Stress also alters brain activity in the visual cortex, which can be monitored using electroencephalography (EEG). This technique involves placing electrodes at multiple scalp locations to capture electrical impulses. Although EEG provides valuable insights, its bulky and intrusive setup limits its practicality in everyday clinical or wearable applications[15]. This use of ML for analyzing complex physiological data is a growing field. Recent studies have demonstrated the power of advanced optimization and deep learning algorithms for diagnostic purposes in medical imaging, such as oral cancer detection, which shares similar challenges of high-stakes classification[8]. Similarly, adaptive optimization algorithms have been successfully used for feature selection in complex medical datasets, enhancing diagnostic accuracy.

Voice analysis has emerged as a critical tool in emotion and expression recognition. Several researchers have emphasized the role of vocal patterns in detecting psychological states[16]. Technologies for vocal network monitoring have been introduced, capable of analyzing ambient sounds that convey emotions such as fear, pressure, or compassion[17]. While keyword-triggered alarm systems exist, current research aims to develop devices that activate autonomously, without requiring user input.

Voice-enabled smart devices such as Amazon Alexa, Apple Siri HomePod, and Google Echo Dot—have gained widespread popularity. However, this study focuses on designing a portable control system tailored for IPV survivors, utilizing smartphones equipped with microphones to perform voice-based monitoring and alert functions.

Modern mobile devices are increasingly capable of evaluating clinical parameters in vulnerable populations. Embedded sensors such as accelerometers, gyroscopes, and fitness trackers allow for the continuous collection of biometric and behavioral data[18]. While these tools are often used recreationally, their potential for targeted surveillance and health monitoring is significant.

Expression recognition, a relatively recent field in computer science, has been under active development for the past two decades[19]. It involves analyzing complex emotional states and applying data mining techniques to create adaptive software that responds to users' psychological needs.

The proposed smart security solution for women and children integrates wearable IoT systems composed of multiple modules, each designed to fulfill specific safety functions. These advanced devices combine sensors, communication interfaces, and intelligent software to enable real-time monitoring, alert generation, and emergency response. The system is engineered to provide continuous connectivity, rapid assistance during crises, and comprehensive surveillance of the wearer's physiological and environmental conditions—ultimately offering peace of mind and enhanced personal security[20,21].

Child safety was the central focus of Vidhya Prasanth et al.'s[22] study titled "*Analysis of Child Safety Monitoring System using the SPSS Method*." This research explored the development of a smart device capable of delivering accurate, real-time location tracking to enhance child protection. The system incorporates features such as a distress alarm, SOS light, and ambient temperature sensor to assist in locating children and ensuring their well-being. Emphasis was placed on an intuitive user interface and the integration of GPS-enabled smartphones for both parents and children, enabling continuous monitoring of movement and vital health parameters including blood oxygen levels, temperature, humidity, and heart rate. The system also provides instant access to emergency services, empowering parents to make swift decisions during critical situations.

Similarly, Thummalakunta et al.[23] proposed the "Implementation of IoT-based Real-time Women's Safety System," which introduces a portable device designed to enhance women's security. Upon activation, the device emits a loud buzzer to deter potential attackers and facilitate escape. Beyond immediate protection, the system supports group messaging to a predefined network of contacts—including family members and law enforcement—during emergencies. The study underscores the importance of real-time communication and location awareness for timely intervention. By fostering community engagement and promoting collective action against social threats, this IoT-driven solution aims to transcend conventional personal safety tools and serve as a platform for empowerment and social resilience.

While both Prasanth et al.[22] and Thummalakunta et al.[23] highlight the functional benefits of their IoT-based safety systems, their studies offer limited insight into the security and privacy mechanisms embedded within these frameworks. Specifically, the literature does not elaborate on how sensitive personal data—such as location, health metrics, and emergency communications—is protected from unauthorized access or misuse.

Prasanth et al.'s[22] system places significant emphasis on user-friendly design and real-time health monitoring. However, the reliance on GPS and smartphone integration introduces potential vulnerabilities related to data privacy. The study does not detail the safeguards implemented to protect user data throughout its lifecycle—from collection and transmission to storage and access by guardians or emergency personnel.

This review complements earlier discussions on technological interventions for survivors of Intimate Partner Violence (IPV), particularly those involving biosensors and wearable devices for autonomous threat detection and alert generation. While these innovations demonstrate promising capabilities, the absence of robust privacy protocols and data protection strategies remains a critical gap in current research—one that must be addressed to ensure ethical deployment and user trust in safety technologies.

Patient-focused healthcare applications are important to patients because they offer a range of advantages that add value and improve the overall healthcare experience. The 5G networks, along with Mobile Edge Computing (MEC), can greatly transform healthcare applications, which in turn improves patient care. MEC plays an important role in the healthcare of patients by bringing computing resources to the edge of the network. It becomes part of an IoT system within healthcare that brings data closer to the core, speeds up decision-making, lowers latency, and improves the overall quality of care[24–26].

Summary of key literature contributions

- **Hany et al.**[2]: Proposed a multi-modal approach to hostility detection by monitoring electrodermal activity, electroencephalogram (EEG), and electrocardiogram (ECG) signals. The study also suggested incorporating additional physiological characteristics to enhance detection accuracy.
- **Patel and Hassan**[3]: Introduced a wearable device equipped with an integrated altimeter to detect hostile behavior. However, the system's detection capabilities were limited to the sensor's range and lacked broader contextual analysis.
- **Goodwin**: Developed a violence detection model combining cardiovascular metrics, existential indicators, and motion data via accelerometry. The study focused specifically on individuals with Down syndrome, limiting its applicability to general IPV scenarios.
- **Wu et al.**[5]: Demonstrated that systems capable of accurately monitoring daily activities can serve as indicators of a person's independence and functional capacity, offering potential for behavioral analysis in safety applications.

- **Body Area Networks (BANs)**: Initially proposed by Broens et al.[6] for asthma treatment, BANs utilize 24-hour monitoring to anticipate health events. Subsequent enhancements by other researchers[7] introduced data management capabilities. BANs rely on continuous operation of wearable tracking systems equipped with communication modules, accelerometers, and GPS.
- **Emotion Analysis**[8]: Employs respiration patterns and distress signals—such as crying or laughing—to establish physiological trends and assess emotional states, contributing to real-time behavioural monitoring.
- **Breath Detectors**[9]: Offer non-invasive, personalized tracking of health variables. These sensors show promise in disease diagnosis and monitoring individuals at risk of IPV, although their clinical adoption remains limited.
- **Heart Rate (HR) and Photoplethysmography**[11]: Widely used in wearable technology for non-invasive pulse monitoring. While ECG captures electrical signals, photoplethysmography measures blood flow using light. Elevated heart rate due to chronic stress is a medically recognized indicator of psychological distress.
- **Glycaemia**[12]: Variations in glucose levels, stress-related hormonal imbalances, and elevated blood oxaloacetate—often linked to obesity—are being explored as potential biomarkers for violence risk.
- **Skin Conductance (SC)**[13,14]: Measures the skin's electrical activity to assess emotional arousal. Lower SC readings correlate with reduced excitement. Affordable devices like Empatica E4, Microsoft Band 2, and the eHealth Sensor Platform have demonstrated effective SC measurement capabilities.
- **EEG**[15]: Captures stress-induced changes in brain impulses, particularly in the visual cortex. Despite its diagnostic value, EEG remains invasive due to its bulky equipment, limiting its practicality in wearable applications.
- **Voice Analysis**[16,17]: Recognized as a powerful tool for emotion detection. Technologies have emerged to monitor vocal patterns and ambient sounds that convey fear, pressure, or compassion. The goal is to develop autonomous alert systems that do not rely on manual activation, leveraging widely available smart devices such as Amazon Alexa, Apple Siri HomePod, and Google Echo Dot.
- **Mobile Devices**[18]: Ubiquitous and capable of evaluating clinical parameters in vulnerable populations. Embedded sensors—including accelerometers, fitness trackers, and gyroscopes—enable continuous health and activity monitoring.
- **Expression Recognition**[19]: A growing field in computer science focused on analysing complex emotional states. It utilizes data mining techniques to develop adaptive systems that respond to users' psychological needs.
- **Smart Security Solutions using Wearable IoT Systems**[20]: Represent a comprehensive approach to enhancing safety for women and children. These systems integrate sensors, communication modules, and intelligent software to enable real-time monitoring, alert generation, and emergency response.

Table 1 presents a chronological summary of significant research contributions over the past two decades, highlighting key developments in the fields of physiological response monitoring, wearable technologies, and IoT-based safety systems. This timeline illustrates the progressive integration of biometric sensing, machine learning, and real-time communication technologies into personal safety frameworks. Each entry includes the author(s), publication venue, and a brief description of the study's core innovation, offering a clear view of how the field has evolved to address emerging safety challenges.

Table 2 provides a comparative analysis of commonly used IoT hardware components in safety and security applications. It outlines the strengths of each device—such as versatility, high accuracy, and real-time data processing capabilities—while also acknowledging their limitations. These include restricted computational power, elevated energy consumption, and potential vulnerabilities related to data privacy and security. The table serves as a practical reference for selecting appropriate hardware platforms based on specific application requirements and operational constraints. Furthermore, the efficiency of these ML models is a key area of research. Recent work has focused on hybrid metaheuristic optimization and innovative feature selection methods to build more accurate and computationally efficient models, which is critical for real-time IoT applications.

However, a comprehensive review of the existing literature reveals a persistent gap: while many systems focus on either women or children, few provide a unified framework that addresses their interconnected vulnerabilities. Moreover, many existing datasets and models lack consideration for cultural diversity and demographic fairness, which can lead to significant biases in stress and emotion recognition. This oversight, combined with a reliance on outdated, reactive technologies, forms the primary motivation for the research gaps detailed in the following section.

Research gaps

Despite notable advancements in modern safety technologies, substantial gaps persist in safeguarding survivors of Intimate Partner Violence (IPV), particularly among women and children. While many contemporary systems utilize machine learning (ML) algorithms to autonomously detect threats and initiate alerts—eliminating the need for manual activation—these solutions still fall short in delivering comprehensive, context-aware protection. Autonomous detection improves responsiveness and reduces false alarms through controlled deactivation protocols, yet broader infrastructural and collaborative efforts are essential to support diverse user needs and ensure practical deployment.

This research extends the capabilities of wearable safety devices by introducing intelligent, real-time, context-sensitive systems. However, existing literature and technological implementations often treat women's and children's safety as distinct domains, overlooking their interconnected vulnerabilities. Furthermore, many

Year	Authors	Title/Focus	Publication Venue	Key Contribution
2001	Grossman & Wientjes	Breathing adjustments to mental/physical demands	Springer	Foundational work on respiratory response to stress[8]
2002	Rand et al.	Acute stress hyperglycemia in stress response	J. Vet. Intern. Med.	Physiological markers of acute stress[12]
2007	Broens et al.	Context-aware m-health application comprehensive framework	Int. J. Internet Protocol Technol.	Early mobile health comprehensive framework[6]
2010	Yuce	Wireless body area networks for healthcare	Sens. Actuators A Phys.	WBAN implementation foundations[7]
2010	Picard	Affective Computing developments	IEEE Trans. Affect. Comput.	Emotion recognition computing[19]
2013	Brouwer et al.	Emotional stimuli effects on physiological variables	Front. Hum. Neurosci.	Physiological emotion correlation[13]
2013	Szwoch	Physiological signals for emotion recognition	Int. Conf. Human System Interaction	Signal processing for emotions[16]
2015	Case et al.	Accuracy of smartphone/wearable activity tracking	JAMA	Validation of consumer devices[18]
2016	Jatti et al.	IoT wearable device for women's safety	IEEE RTEICT	Early IoT safety device[4]
2017	Kumari et al.	Wearables and multimodal interfaces trend	Biosens. Bioelectron.	Comprehensive wearables review[28]
2018	Patel & Hasan	Smart bracelets for personal safety automation	IEEE CCNC	Smart jewelry for safety[3]
2018	Islam et al.	SafeBand: Wearable device for women in Bangladesh	Mobile Computing Conference	Regional safety solution[5]
2018	Kutt et al.	Sensor platform for physiological data processing	Int. Conf. AI and Soft Computing	Data processing architecture[14]
2018	Byrom et al.	Brain monitoring devices in clinical research	Clin. Pharmacol. Ther.	Remote monitoring potential[15]
2018	Joy	Voice assistants introduction	Med. Ref. Serv. Q.	Voice interface integration[29]
2019	Ferdinando et al.	VITEC: Violence Detection Comprehensive framework	AICON Conference	Violence detection system[2]
2019	Goodwin et al.	Predicting aggression in autism using biosensors	Autism Res.	Behavioral prediction[27]
2019	Güntner et al.	Breath sensors for health monitoring	ACS Sens.	Respiratory monitoring technology[9]
2019	Haghayegh et al.	Fitbit accuracy for sleep heart rate assessment	Chronobiol. Int.	Consumer device validation[10]
2019	Sumathy et al.	Virtual Friendly Device for Women Security	Physics Conference	Virtual safety assistance[17]
2019	Rachakonda et al.	Stress-Lysis: DNN-integrated edge device	IEEE Trans. Conum. Electron.	Edge computing for stress detection[30]
2020	Rodríguez-Rodríguez et al.	Holistic ICT platform for IPV survivors	Symmetry	Comprehensive IPV protection[1]
2024	Wagh et al.	Smart Security Solution using Wearable IOT	Wireless Pers Commun	Recent IoT security advancement[20]
2024	Prasanth et al.	Child Safety Monitoring System analysis	IJEAST	Child-focused safety systems[22]
2024	Thummalakunta et al.	Real-time Women's Safety System implementation	IJERT	Real-time safety implementation[23]

Table 1. Summary of literature survey in chronological order researcher represented work[8,20,27].

Safety & security harware devices	Advantages	Disadvantages
Microcontrollers (Arduino, Raspberry Pi)	Versatile: Can be programmed for various tasks; Cost-effective: Relatively inexpensive; Open-source: Large community support and resources; Compact: Suitable for wearable or embedded applications[1,2,4,27]	Limited Processing Power: May struggle with complex AI/ML tasks; Power Consumption: Can be higher than specialized chips; Learning Curve: Requires programming knowledge[1,2,4,27]
GPS Modules	Accurate Location Tracking: Essential for emergency services; Global Coverage: Works in most outdoor areas; Real-time Navigation: Provides turn-by-turn directions[2–4]	Power Intensive: Can drain battery quickly; Indoor Limitations: Poor performance indoors/underground; Privacy Concerns: Continuous tracking issues[2–4]
Accelerometer/Gyroscope	Movement Detection: Detects falls, impacts, orientation changes; Low Power Consumption: Generally energy efficient; Real-time Response: Instant motion sensing[3,4,27]	False Positives: Can trigger during normal vigorous activity; Calibration Required: Needs tuning for different users; Environmental Sensitivity: Affected by temperature and vibration[3,4,27]
Camera Module	Visual Evidence: Captures footage for security/prosecution; Contextual Information: Provides visual context of incidents; Remote Monitoring: Live streaming capabilities[1–4,27]	High Power Consumption: Significant battery drain; Privacy Concerns: Major privacy implications; Storage Requirements: Needs substantial storage space; Legal Implications: Recording laws vary by jurisdiction[1–4,27]

Table 2. Summary of IoT devices with advantages and Disadvantages[1–3,27].

systems continue to rely on outdated mechanisms such as manual reporting, SMS alerts, and basic GPS tracking, which limit accuracy, reliability, and responsiveness during emergencies.

Technical research gaps

1. Lack of Integrated IoT-Based Systems: Current systems lack unified architectures capable of simultaneously monitoring and analyzing the safety of both women and children in real time.
2. Limited Application of Machine Learning: The full potential of ML algorithms in predicting and mitigating safety risks remains underexplored.
3. Underutilization of Sensor Data: Existing solutions fail to effectively harness data from wearable devices, environmental sensors, and digital platforms such as social media.
4. Insufficient Context-Awareness: Many systems lack the ability to interpret contextual variables, which are essential for delivering personalized alerts and recommendations.

5. Scalability and Interoperability Challenges: Technical limitations hinder large-scale deployment and seamless integration across platforms and devices.

Social and cultural research gaps

1. **Lack of Cultural Sensitivity**: Regional cultural and social dynamics are often overlooked, reducing the relevance and effectiveness of safety systems.
2. **Minimal Stakeholder Engagement**: Limited engagement with key stakeholders—including advocacy groups, child welfare organizations, and law enforcement—impairs system design and adoption.
3. **Low Awareness and Education**: There is a need for targeted campaigns and educational initiatives to inform users about available safety technologies and services.

Methodological research gaps

1. **Absence of Standardized Evaluation Metrics**: A consistent framework for assessing the performance and impact of IoT-based safety systems is lacking.
2. **Limited Use of Mixed-Methods Approaches**: Few studies integrate both qualitative and quantitative approaches to provide a holistic understanding of system effectiveness.
3. **Neglect of Ethical Considerations**: Critical issues such as data privacy, security, and informed consent are often inadequately addressed in current research.

By identifying and addressing these gaps, this study contributes to the development of a robust, intelligent, and ethically responsible IoT-based safety system for women and children—powered by advanced machine learning techniques and designed for real-world deployment.

Research challenges

The development and deployment of safety solutions for women and children using Machine Learning (ML), sensor technologies, and Internet of Things (IoT) systems face a range of multifaceted challenges. These span technical, methodological, social, and infrastructural domains, each presenting unique barriers to effective implementation and scalability:

Technical challenges

1. Data Availability and Quality: Incomplete or low-quality datasets can compromise the accuracy of ML models, leading to missed detections or false alarms.
2. Sensor Maintenance and Calibration: Regular upkeep and calibration of sensors are essential but can be costly and labor-intensive, affecting long-term sustainability.
3. Interoperability Problems: Integrating heterogeneous sensors, IoT devices, and ML algorithms often leads to compatibility challenges and system inefficiencies.
4. Scalability and Flexibility: Existing solutions may lack adaptability to emerging technologies or evolving user requirements, limiting their long-term relevance.
5. Battery Life and Power Consumption: Short battery lifespans in IoT devices necessitate frequent charging or replacement, which can hinder continuous operation.

Machine learning challenges

1. Class Imbalance: ML models may struggle when safe scenarios vastly outnumber risky ones, reducing the system's ability to detect threats accurately.
2. Overfitting and Underfitting: Poor generalization can result from models that either overlearn training data or fail to capture meaningful patterns.
3. Adversarial Attacks: ML systems are susceptible to adversarial attacks that can manipulate inputs and degrade performance.
4. Lack of Explainability: Many ML models operate as "black boxes," making it difficult to interpret decision-making processes and build user trust.
5. Real-Time Processing Limitations: Delays in data processing can impede timely responses during emergencies, undermining system effectiveness.

IOT and sensor challenges

1. Accuracy and Reliability of Sensors: Inconsistent sensor readings can lead to false positives or missed alerts, compromising user safety.
2. Sensor Orientation and location: The effectiveness of sensors is highly dependent on their positioning, which can vary across users and environments.
3. Wireless Communication Problems: Connectivity issues between devices may result in dropped signals or delayed transmissions.
4. Security and Privacy Risks: IoT systems often collect sensitive personal data, raising concerns about unauthorized access and data misuse.
5. Environmental conditions: External factors such as weather, lighting, and noise can affect sensor performance and data accuracy.

Social and cultural challenges

1. Social and Cultural Barriers: Societal norms and stigmas may hinder the adoption and effectiveness of safety technologies.
2. Lack of Knowledge and Instruction: A lack of education and outreach prevents many users from understanding or accessing available safety solutions.
3. Stigma and Shame: Victims may hesitate to report abuse or use safety tools due to fear of judgment or social repercussions.
4. Power Dynamics: Unequal power relationships can influence the design and implementation of safety measures, affecting their accessibility and fairness.
5. Stakeholder Involvement: Minimal involvement from key stakeholders—such as law enforcement, healthcare providers, and advocacy groups—can limit system impact.

Economic and infrastructure challenges

1. Cost and Affordability: High costs may prevent vulnerable populations from accessing advanced safety technologies.
2. Infrastructure Limitations: Poor lighting, limited electric supplies, limited mobility, and inadequate internet connectivity can reduce system effectiveness in certain regions.
3. Resource Constraints: The efficacy of safety solutions may be impacted by infrastructure constraints including inadequate mobility, bad lighting, or a lack of internet connectivity.
4. Policy and Regulatory Comprehensive frameworks: The absence of supportive legal and regulatory frameworks may delay adoption and reduce system credibility.
5. Public-Private Partnerships: Insufficient partnerships between government and private sectors can restrict funding and innovation in safety technology development.

By addressing these challenges, this research aims to pave the way for more resilient, inclusive, and ethically grounded safety systems that protect women and children in diverse environments.

Figure 1 presents a tree diagram illustrating the key inadequacies in current IoT systems designed for women and child safety. The tree structure uses a central trunk with branches extending outward, each representing a different systemic problem.

A conceptual diagram highlights six critical shortcomings in current IoT-based safety systems, particularly those designed for women and children. These foundational issues must be addressed to develop more effective, integrated, and responsive safety technologies. The analysis is framed within a "protection shield" model—represented in red—to emphasize the urgency of resolving these challenges.

Limited Scope - Most existing systems are tailored exclusively for either women or children, lacking a unified framework that addresses the safety needs of both demographics in a cohesive manner.

Reactive Systems - Many solutions are built to respond only after an incident has occurred, often requiring manual activation. This reactive approach diminishes their ability to prevent harm or deliver immediate assistance during emergencies.

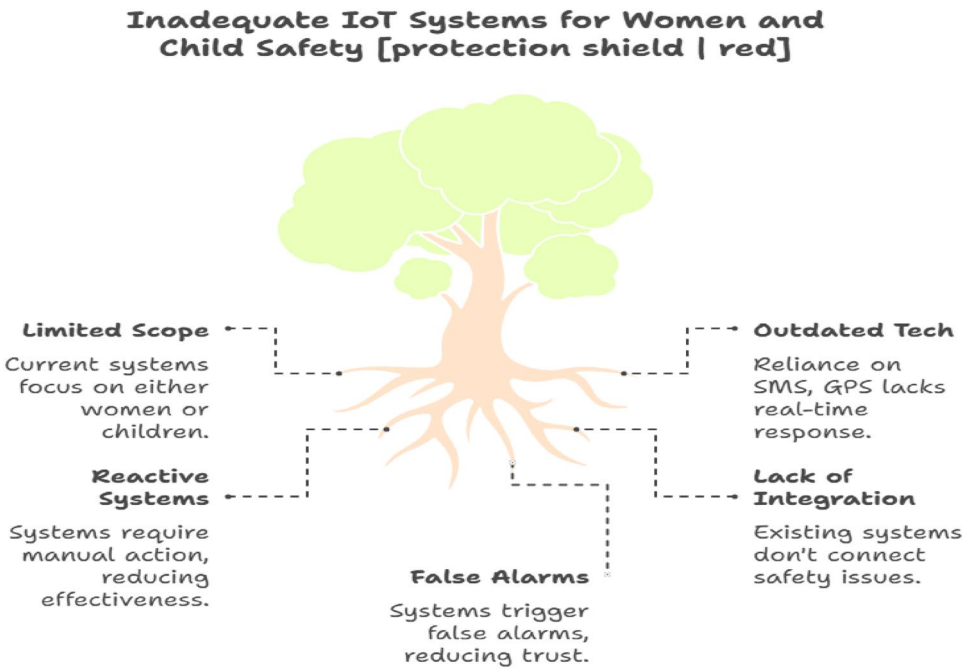

Fig. 1. Inadequate IoT Systems for Women and Children Safety.

Outdated Technology - Reliance on legacy tools such as SMS and basic GPS tracking limits the systems' responsiveness and precision. These technologies are inadequate for real-time threat detection and rapid intervention.

Lack of Integration - Safety devices often function in isolation, without connectivity to broader safety infrastructure such as emergency services, smart city networks, or healthcare systems. This fragmentation reduces the potential for coordinated and comprehensive protection.

False Alarms - Frequent false alerts undermine user confidence and may lead to desensitization or delayed responses during actual emergencies. Enhancing detection accuracy is essential to maintain trust and system reliability.

The diagram employs a tree metaphor to illustrate that these issues are not superficial but deeply rooted in the design and implementation of current safety systems. Addressing these "root-level" problems is essential for cultivating robust, intelligent, and inclusive IoT safety solutions that serve both women and children effectively.

Problem statement

The rising incidence of crimes targeting women and children underscores the urgent need for reliable and intelligent safety solutions. However, existing systems suffer from a critical design flaw: they are overwhelmingly reactive. Many solutions rely on manual activation, such as pressing a button, which is often impossible when a victim is incapacitated or under duress. To overcome this, some systems have attempted autonomous detection by monitoring physiological parameters like body temperature and heart rate.

However, these systems fall into a second critical trap: a high rate of false positives. They often use simple, static thresholds to detect distress. This oversimplification fails to account for real-world variability; for instance, a user's heart rate will naturally elevate during exercise, and this benign activity could be misidentified as a threat[cite: 306]. Such false alarms severely erode user trust and render the system unreliable. Therefore, the core problem is not just to detect physiological anomalies, but to distinguish between true distress and benign, everyday activities.

This research addresses this specific problem by developing a personalized, context-aware wearable safety device. Instead of relying on generic, fixed thresholds, our system uses a machine learning (ML) model trained on sensor data (including heart rate, temperature, and motion) to learn the user's individual, baseline patterns [cite: 307, 310]. The system is designed to autonomously identify significant deviations from this personal baseline, fusing data from multiple sensors to validate a potential threat before triggering an alert. This sensor fusion and ML-based anomaly detection approach is designed to minimize false positives and provide a reliable, autonomous emergency notification system, complete with real-time GPS location, for rapid intervention[cite: 308].

Dataset To support the development and validation of the proposed system, three datasets were utilized.

1. **Stress-Lysis**: This dataset provides 2001 samples of human body humidity, body temperature, and step count. It is used to train the model to perform three-class classification of stress levels: Low, Normal, and High, based on physical activity [cite: 314–316].
2. **Smart Phone Dataset for HAR**: This dataset was collected from 30 participants (aged 22–79) using smartphone accelerometers and gyroscopes.It provides labeled data for six common activities (standing, sitting, laying, walking, walking upstairs, walking downstairs), which is essential for training the model to recognize a user's activity context.
3. **IoT Wearables Dataset for Women's Safety: Stress Detection and Analysis** .

The enhanced dataset is a sophisticated collection of simulated data points, meticulously designed to emulate real-world data as collected from wearable Internet of Things (IoT) devices. This dataset is tailored for applications in safety monitoring, particularly for women, and is ideal for developing machine learning models for distress or danger detection. Number of Samples: 1000 entries[31].

Proposed system design

System architecture and deployment

The proposed system is a wearable, intelligent safety device designed for continuous, autonomous monitoring. The architecture, illustrated in Fig. 1, is a closed-loop system centered on an Arduino microcontroller, which acts as the central processing unit for data acquisition, and a Raspberry Pi 3, which handles more intensive machine learning (ML) inference. The system's logic follows a clear data pipeline:

(1). **Data Acquisition**: Physiological sensors (Heart Rate, Temperature Sensor) and environmental sensors (GPS, Accelerometer, Tilt Sensor) continuously gather real-time data from the user.
(2). **Edge Processing**: This raw data is fed to the microcontrollers (Arduino and Raspberry Pi 3) for on-device, or "edge," processing. This is critical for minimizing latency, as it eliminates the need for a constant, high-speed connection to a cloud server. The ML models (SVM, NB) are deployed directly on the Raspberry Pi.
(3). **AI-Driven Inference**: The ML model analyzes the fused sensor data in real-time, comparing it against the user's trained personal baseline to detect significant anomalies indicative of distress.
(4). **Action & Alerting**:

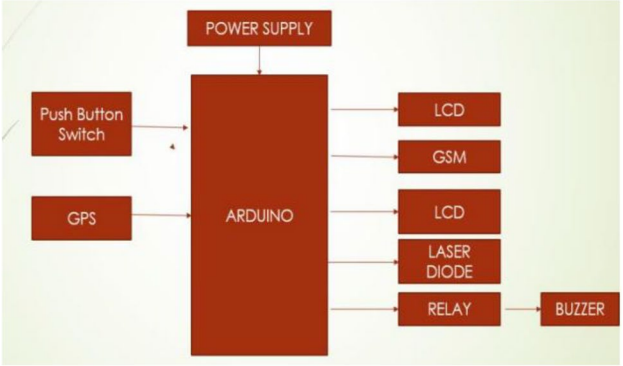

Fig. 2. Proposed system design.

- **Autonomous Alert**: If the ML model classifies a high-threat event, the system automatically triggers an alert.
- **Manual Alert**: The user can also manually trigger an alert using the Push Button Switch.
- **Response**: Upon an alert, the Arduino activates the local Buzzer (via Relay) and Laser Diode to create a local deterrent. Simultaneously, the GSM module retrieves the user's precise location from the GPS module and transmits an emergency alert with coordinates to pre-registered contacts and a secure endpoint for law enforcement.

Computational and deployment considerations

This edge-computing architecture is essential for practical, real-world deployment. The SVM model, in particular, is computationally lightweight, making it ideal for running on the Raspberry Pi 3 with minimal processing delay (achieving detection in ~3 s, as shown in our results). For integration with law enforcement, the GSM alert is formatted as a JSON data packet sent via a secure GPRS connection to a dedicated API endpoint. This allows the alert to be fed directly into a computer-aided dispatch (CAD) system, providing officers with the victim's location in real-time. Power management is optimized to provide 18–20 h of continuous operation, ensuring the device remains active throughout a user's day.

The proposed system design is illustrated in Fig. 2.

Hardware components

Arduino

The latest version of the Arduino Uno includes a USB interface, six analog input pins, and fourteen digital I/O connectors for connecting to external electrical components. Among the 14 I/O ports, six can be used for pulse width modulation (PWM) output.

LCD

This type of electronic screen is used in a multitude of devices, including cell phones, laptops, television sets, and more. These screens often use diodes that produce light in multiple segment and seven-segment configurations. There are several advantages to using this component, such as its affordability, easy configurability, and the ability to display animations and other visuals without restrictions.

Switch

A pushbutton is a simple switch used to control certain components of an appliance or operation. Typically, these buttons are made from rigid materials, most often metal or plastic.

Buzzer

Buzzers are audio signaling devices that can be electronic, mechanical, or piezoelectric. They are commonly used for various purposes, including alarm clocks, reminders, and confirming user input, such as pressing a mouse or keyboard button. Buzzers are mechanical transducers powered by direct current and are frequently utilized in audio devices like laptops, printers, photocopiers, timers, mechanical toys, automotive electronics, cell phones, clocks, and other technological devices.

Power supply

The power supply delivers electrical power to various components, typically those attached to the driven segment. In computers, power supplies, often located at the rear of the case, include at least one fan. These components convert alternating current (AC) to direct current (DC).

Temperature sensor

Given the critical importance of maintaining one's internal body temperature for health, it is essential to monitor it consistently. Various temperature sensors can accurately determine body temperature. A prime example of a highly accurate circuit-based detector is the LM35 series. This sensor produces a voltage output linearly correlated to the temperature in Celsius, operating with an indication factor of ± 10mV/°C and a precision of

0.5 °C. Significant fluctuations in body temperature during critical situations can activate emergency response mechanisms[28].

Pulse rate sensor
The heart beat sensor provides a computerized output of heartbeats. When functioning correctly, the LED lights up with each heartbeat. To determine the beats per minute (BPM) rate, the electronic output is directly linked to a microprocessor. The sensor's operation is based on the light flashing of linked satellites, which are monitored to uplink information and achieve synchrony. The entire system utilizes the L band, spanning from 1.2 to 1.6 GHz, and operates across four different frequencies.

GSM
GSM-compatible SIM cards can transmit location information obtained via GPS. The phone number stored on the GSM SIM card is linked to the system's database. In this proposed device, GPS acts as the transmitter, while GSM functions as the receiver. The parameters from the signaling device are sent as a text message to preset emergency phone numbers. The connection between the 16th port of the microprocessor and the GSM system's reception pin facilitates this process. The voltage regulator circuit supplies an input voltage of approximately 3.4 volts or higher.

GPS
This device is used for routing and precise positioning. It records location coordinates in terms of longitude and latitude. A GPS device receives information from all GPS satellites, including the exact time signals are transmitted. By subtracting the time, the signal was sent from the time it was received, the GPS can calculate the distance to each satellite.

Audio classifier
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Laser diode
A semiconductor component called a laser diode produces high-intensity coherent illumination. Laser diodes operate on the principle of stimulated emission. While similar to light-emitting diodes (LEDs), laser diodes generate coherent radio waves at the photon junction. This means that the electromagnetic waves produced have the same wavelength and phase, resulting in what is known as "coherent radiation."

This integrated system design combines sensor fusion, machine learning, and real-time communication to deliver a comprehensive safety solution tailored for women and children. It ensures continuous monitoring, accurate threat detection, and immediate emergency response—contributing to a safer and more empowered society.

Methodology

The methodology for this system is twofold: first, the development and robust validation of the machine learning models, and second, the design of the system's operational and security framework.

For the ML models, data from the specified datasets was preprocessed and standardized. To ensure the models are robust and to prevent overfitting, a 10-fold cross-validation technique was employed. The dataset was partitioned into 10 subsets; the models were trained on 9 folds and tested on the remaining fold, with this process repeated 10 times. The final reported accuracy represents the average of these folds. This rigorous validation is critical for confirming the high accuracy of the SVM and Naive Bayes classifiers. For the Deep Learning (CNN) model, additional overfitting mitigation techniques, such as Dropout layers and Early Stopping (where training is halted if validation accuracy does not improve), were implemented to ensure the model generalizes well to unseen data.

The proposed IoT-based safety system for women and children is structured around four core operational mechanisms: Manual, Automated, Alert-Based, and Security & Privacy. Each mechanism is designed to ensure responsive, reliable, and ethically sound protection in real-time scenarios.

1. **Manual Mechanism**: In situations where the user is conscious and able to act, a manual trigger is available. By pressing a designated switch, the system activates a loud buzzer to alert nearby individuals and draw immediate attention. This action also initiates the alert protocol, enabling further emergency communication.
2. **Automated Mechanism**: Recognizing that users may be incapacitated or unable to manually respond, the system incorporates automated detection using multiple sensors—including temperature, heart rate, and abrasion sensors. Data from these sensors is cross-validated to minimize false positives. If two or more sensors detect anomalies, the security protocol is activated.

A force-sensitive resistor (FSR) is employed to measure pressure changes. Even slight increases in force result in significant impedance reduction, which is converted into an analog voltage ranging from 0 to 5 centivolts. This sensor fusion approach enhances reliability and responsiveness without requiring user input.

3. **Alert-Based Mechanism**: Upon detection of a potential threat—whether manually or automatically triggered—the system activates its alert mechanism. Using GPS and GSM technologies, it sends real-time loca-

tion data via SMS to pre-registered emergency contacts and local authorities. The location is embedded with Google Maps links for rapid access and navigation.

Security and privacy comprehensive framework for iot-based women and child safety system

Given the sensitive nature of the data collected—such as biometric readings, GPS coordinates, and stress indicators—the system incorporates a robust, multi-layered security and privacy architecture to safeguard user information.

Key features include:

- End-to-End Encryption (AES-256): All data transfers between wearable sensors, microcontroller units (Arduino/Raspberry Pi), and cloud-based ML platforms are encrypted to prevent unauthorized access.
- Data Minimization and Edge Processing: Only essential biometric parameters are collected. Wherever possible, data is processed locally on edge devices to reduce cloud exposure and enhance privacy.
- Differential Privacy in ML Models: Algorithms are designed to preserve model accuracy while anonymizing user-specific patterns, ensuring ethical data handling.
- Granular Consent Management: Users can control geographic tracking boundaries, emergency contact permissions, and data-sharing preferences through multi-tiered consent protocols.
- Role-Based Access Control: Only verified family members, designated emergency contacts, and authorized law enforcement personnel can access sensitive data during confirmed emergencies.
- Secure Boot and Firmware Protection: The system prevents unauthorized firmware manipulation through secure boot mechanisms and encrypted update channels.
- Blockchain-Based Audit Trails: All emergency alerts and data access events are logged using blockchain technology to ensure transparency and accountability.
- Automatic Data Retention Policies: Configurable purge schedules ensure that historical biometric and location data is not retained indefinitely, minimizing long-term privacy risks.
- Adversarial Defense Mechanisms: Input validation, anomaly detection, and federated learning are employed to protect the system from adversarial attacks and maintain model integrity without centralized data exposure.
- This comprehensive methodology ensures that the safety system remains responsive, secure, and privacy-conscious—delivering reliable protection for vulnerable users in both everyday and emergency scenarios.

Multilayer architecture

Figure 3 describe provide multilayer architecture which generate Alert. The Alerting System's escalate with deviation severity,

1. Personalized Baselines: The system creates individual baseline patterns over a period of 7–14 days in place of generic population averages;

- • Multi-tier Architecture: Three warning levels (Caution, Alert, and Emergency) based on statistical departures from personal norms.
- • Adjustable Calculation: Employs statistical techniques that incorporate severity multipliers and environmental considerations.

2. Fusion data from multiple sensors.

Weighted integration combines GPS data, motion, skin conductance, temperature, and heart rate. Advanced filtering (Kalman filters) and feature extraction techniques are used in signal processing. Individual sensor analysis and cross-sensor validation are the two levels of the 3.

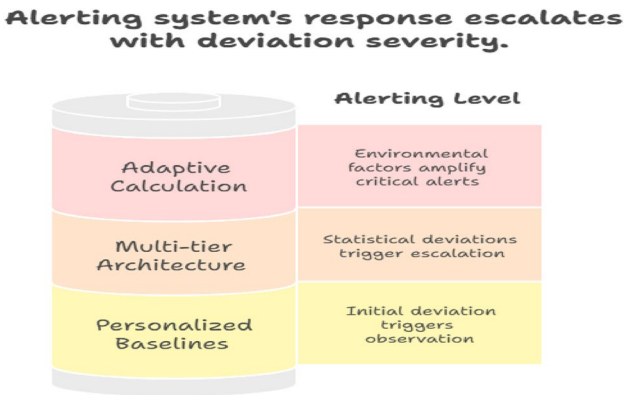

Fig. 3. Alerting System's Response Escalates with Deviation severity[32].

3. Hierarchical Fusion technique.
Emergency Classification Logic.

- Machine Learning Pipeline: Utilizes the SVM and Naive Bayes results.
- Context Awareness: Factors in location, time of day, and activity patterns.
- Pattern Recognition: Distinguishes between panic attacks, physical assault, medical emergencies, and false alarms.

4. Key Technical Innovations.

- Minimum Agreement Rule: Requires at least 2 sensors to agree before triggering alerts.
- Temporal Consistency: Anomalies must persist for specific durations to avoid false positives.
- Continuous Learning: System adapts based on user feedback and population-level patterns.
- *Signature Method of Connected Devices*:

IoT devices are primarily signed using digital signatures based on Public Key Infrastructure (PKI), which cryptographically ties a device's identity to its data to guarantee non-repudiation, authenticity, and integrity. In this process, a device uses its own private key to sign a hash of its data, which can be validated by anybody with the matching public key. This ensures that the data is authentic and hasn't been altered. Furthermore, new uses of blockchain technology improve device authentication and data provenance by offering decentralized and unchangeable identity management for linked devices. Secure Boot mechanisms also use digital signatures to confirm the integrity of a device's firmware and operating system from the first power-on, preventing unauthorized code execution.

In the context of IoT-based women and children's safety, sensor thresholds are dynamically determined through continuous data analysis and machine learning.

- **Normal Situations**: Large datasets of typical user behavior and ambient situations are used to train the AI/ML models. In doing so, they are able to create a baseline for "routine activities." For example, a regular scenario could include an accelerometer readout that indicates walking or sitting, a heart rate that falls within a normal range, and ambient noise levels typical of an office or public setting. The system picks up on the minute differences and trends of these typical states.
- **Emergency Situations**: Deviations from these learned normal patterns, especially when multiple sensor inputs are fused, trigger the classification of an event as a potential emergency. For example, a sudden, sharp increase in heart rate combined with erratic accelerometer data (suggesting a fall or struggle), and a sudden loud noise or a specific vocal pattern (indicating distress), would collectively exceed the dynamically set thresholds, signaling an emergency. The AI/ML algorithms are crucial in identifying these complex, multi-modal anomalies that distinguish a genuine threat from a mere benign anomaly.

Discussion on power and maintenance of wearable devices

1. The efficacy and practicality of IoT-based wearable safety devices are significantly influenced by their power management and maintenance requirements. In this study, a key focus was placed on optimizing the device's energy consumption to ensure prolonged operational capability. The implemented battery optimization strategies have successfully extended the usable life of the wearable to 18–20 h, which is crucial for continuous real-time GPS monitoring and GSM-based emergency communication, especially during extended periods of travel or work. This optimized battery performance is vital for user confidence and reliable protection, minimizing the need for frequent recharging and ensuring the device remains active when most needed.
2. Beyond power, the long-term viability and user acceptance of such devices also depend heavily on their ease of maintenance. While not explicitly detailed in the core findings, a robust IoT wearable safety device would necessitate minimal and user-friendly maintenance. This typically includes:

- **Regular Charging**: Users would need to establish a routine for charging the device, similar to a smartphone.
- **Software Updates**: Periodic over-the-air (OTA) software updates would be essential to enhance security features, improve algorithm accuracy (e.g., for refined sensor thresholding and emergency detection), and introduce new functionalities.
- **Physical Upkeep**: Basic physical maintenance, such as keeping the device clean and dry, would contribute to its longevity.
- **Sensor Calibration/Checks**: Although designed for robustness, occasional checks or recalibrations of sensors might be beneficial to ensure continued accuracy, especially after prolonged use or exposure to extreme conditions.
- Designing these devices with durable materials and sealed enclosures would further reduce the need for complex maintenance, making them more resilient to daily wear and tear and various environmental conditions.

Let me describe the functions, advantages, and disadvantages of this type of architecture:

Functions of Multi-Tier alerting architecture

Tier 1 - Personalized baselines (Foundation Layer)

- Establishes individual baseline metrics for each system/user.
- Triggers initial observations when deviations occur.

- Provides the foundational data for higher-tier analysis.

Tier 2 - Multi-tier architecture (Processing Layer)

- Processes statistical deviations and determines escalation needs.
- Implements logic for pattern recognition and anomaly detection.
- Routes alerts to appropriate response channels.

Tier 3 - Adaptive calculation (Intelligence Layer)

- Applies machine learning or adaptive algorithms.
- Amplifies critical alerts based on environmental factors.
- Provides context-aware response scaling.

Advantages

Scalability and efficiency

- Reduces alert fatigue by filtering out minor deviations at lower tiers.
- Allows system resources to focus on genuinely critical issues.
- Supports large-scale monitoring with manageable alert volumes.

Intelligent response management

- Provides graduated response levels rather than binary alerts.
- Adapts to changing conditions and learns from patterns.
- Reduces false positives through multi-layer validation.

Contextual awareness

- Environmental factors enhance decision-making accuracy.
- Personalized baselines improve relevance for individual cases.
- Statistical analysis provides objective deviation assessment.

Disadvantages

Complexity and maintenance

- Requires sophisticated configuration and tuning across multiple tiers.
- Potential single points of failure in the escalation chain.
- Complex troubleshooting when alerts don't escalate properly.

Latency concerns

- Multi-tier processing can introduce delays in critical alert delivery.
- Time spent in lower tiers might delay urgent responses.
- Computational overhead from adaptive calculations.

Risk of Over-Engineering

- May create unnecessary complexity for simpler monitoring needs.
- Potential for missed alerts if tier logic is incorrectly configured.
- Requires ongoing calibration to maintain effectiveness.

This architecture is particularly well-suited for systems requiring nuanced alerting (healthcare monitoring, infrastructure management, security systems) but may be overkill for simpler applications where basic threshold alerting suffices.

Security attacks and solutions for Multi-Tier alerting architecture

Layer 1: personalized baselines

Attack Type	Description	Impact	Solutions
Baseline Poisoning	Attacker manipulates historical data to skew baseline calculations	False baselines lead to missed alerts or excessive false positives	• Implement data integrity checks• Use multiple validation sources• Apply statistical outlier detection during baseline creation• Regular baseline auditing
Data Injection	Malicious insertion of fake data points to corrupt baseline metrics	Compromised baseline accuracy	• Input validation and sanitization• Digital signatures for data sources• Rate limiting on data ingestion• Source authentication
Baseline Drift Attack	Gradual manipulation to slowly shift baseline without detection	Subtle but persistent degradation of alert sensitivity	• Implement baseline change monitoring• Set maximum drift thresholds• Use multiple baseline calculation methods• Regular baseline recalibration
Historical Data Tampering	Modification of stored historical data used for baseline calculation	Corrupted foundation for all alerting decisions	• Immutable data storage (blockchain/append-only logs)• Data backup and versioning• Cryptographic hashing of historical records• Access control and audit logging

Layer 2: Multi-tier architecture

Attack Type	Description	Impact	Solutions
Logic Bomb	Malicious code inserted into processing logic to trigger at specific conditions	System compromise at critical moments	• Code review and static analysis• Runtime behavior monitoring• Sandboxed execution environments• Regular security audits
Escalation Bypass	Circumventing normal tier progression to suppress critical alerts	High-severity incidents go undetected	• Implement multiple escalation paths• Cross-validation between tiers• Independent monitoring of escalation logic• Fail-safe escalation mechanisms
DDoS on Processing	Overwhelming the system with excessive alerts to cause processing delays	System becomes unresponsive during actual emergencies	• Rate limiting and throttling• Load balancing across multiple processors• Priority queuing for critical alerts• Resource isolation
Man-in-the-Middle	Intercepting and modifying alerts between tiers	Altered or blocked critical communications	• End-to-end encryption between tiers• Message authentication codes (MAC)• Certificate pinning• Secure communication protocols
Privilege Escalation	Gaining unauthorized access to higher-tier processing capabilities	Unauthorized control over alert management	• Principle of least privilege• Role-based access control (RBAC)• Regular privilege audits• Multi-factor authentication

Layer 3: adaptive calculation

Attack Type	Description	Impact	Solutions
Model Poisoning	Contaminating training data or model parameters to degrade adaptive performance	Reduced accuracy in critical alert amplification	• Adversarial training techniques • Model validation and testing• Federated learning approaches • Regular model retraining with clean data
Adversarial Examples	Crafted inputs designed to fool machine learning models	Misclassification of critical vs. non-critical events	• Robust model architectures• Input preprocessing and filtering• Ensemble methods• Adversarial detection algorithms
Model Inversion	Reverse-engineering the adaptive model to understand its decision-making	Exposure of sensitive algorithmic logic	• Model obfuscation techniques• Differential privacy• Limited model exposure• Secure enclaves for model execution
Environmental Manipulation	Falsifying environmental factors to influence alert amplification	Incorrect scaling of alert responses	• Multiple independent environmental sensors• Cross-validation of environmental data• Anomaly detection on environmental inputs• Physical security of sensor infrastructure
Algorithm Substitution	Replacing legitimate adaptive algorithms with malicious ones	Complete compromise of intelligent alerting capabilities	• Code signing and verification• Runtime integrity checking• Immutable algorithm deployment• Continuous algorithm validation

Cross-Layer attacks

Attack Type	Description	Impact	Solutions
Coordinated Multi-Layer Attack	Simultaneous attacks across all three layers	Complete system compromise	• Defense in depth strategy• Independent security monitoring for each layer• Cross-layer correlation analysis• Incident response coordination
Supply Chain Attack	Compromise of third-party components used across layers	Widespread system vulnerability	• Vendor security assessments• Software bill of materials (SBOM)• Regular dependency scanning• Secure software development lifecycle
Insider Threat	Malicious actions by authorized personnel with system access	Circumvention of most security controls	• Zero-trust architecture• Behavioral monitoring• Separation of duties• Regular access reviews and psychological screening
Social Engineering	Manipulation of personnel to gain unauthorized system access	Human-factor based system compromise	• Security awareness training• Multi-person authorization for critical changes• Verification procedures• Incident reporting protocols

General security principles

- **Monitoring and Logging**: Comprehensive logging across all layers with tamper-evident storage.
- **Incident Response**: Automated response capabilities with human oversight.
- **Regular Updates**: Continuous security patching and system updates.
- **Redundancy**: Backup systems and failover mechanisms.
- **Testing**: Regular penetration testing and vulnerability assessments.
- **Compliance**: Adherence to relevant security standards and regulations.

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Pseudocode examples for key attacks
Layer 1 attack examples

1. Baseline Poisoning Attack

```
FUNCTION baseline_poisoning_attack():
  // Attacker gains access to historical data store
  target_user = "critical_system_user"
  historical_data = get_historical_data(target_user)
    // Gradually inject false data points
  FOR i = 1 TO poison_duration:
    fake_value = normal_value + (poison_factor * i)
    inject_data_point(target_user, timestamp + i, fake_value)
    sleep(injection_interval)
    // Result: Baseline calculation becomes skewed
  poisoned_baseline = calculate_baseline(target_user)
  // Now real anomalies won't trigger alerts
END FUNCTION
// Defense Implementation
FUNCTION secure_baseline_calculation():
  raw_data = get_historical_data(user_id)
  // Data integrity check
  FOR each data_point IN raw_data:
    IF NOT verify_signature(data_point):
      flag_suspicious_data(data_point)
      CONTINUE
    // Statistical outlier detection
    IF is_statistical_outlier(data_point, raw_data):
      quarantine_data_point(data_point)
  clean_data = remove_flagged_data(raw_data)
  baseline = calculate_robust_baseline(clean_data)
  RETURN baseline
END FUNCTION
```

```
2. Data Injection Attack
FUNCTION data_injection_attack():
  // Attacker exploits weak input validation
  target_endpoint = "data_ingestion_api"
  WHILE attack_active:
    malicious_payload = {
      "user_id": "legitimate_user",
      "timestamp": current_time(),
      "value": generate_fake_metric(),
      "source": "spoofed_sensor" }
        // Bypass weak validation
    inject_data(target_endpoint, malicious_payload)
    sleep(random_interval())
END FUNCTION
// Defense Implementation
FUNCTION secure_data_ingestion():
  INPUT: data_packet
  // Authentication and authorization
  IF NOT authenticate_source(data_packet.source):
    RETURN reject("Unauthorized source")
  // Input validation
  IF NOT validate_schema(data_packet):
    RETURN reject("Invalid data format")
  // Rate limiting
  IF exceeds_rate_limit(data_packet.source):
    RETURN reject("Rate limit exceeded")
  // Digital signature verification
  IF NOT verify_signature(data_packet):
    RETURN reject("Invalid signature")
  // Store with integrity hash
  store_with_hash(data_packet)
  RETURN accept()
END FUNCTION
```

```
Layer 2 Attack Examples
3. Escalation Bypass Attack
FUNCTION escalation_bypass_attack():
  // Attacker compromises escalation logic
  target_alert = get_critical_alert()
  // Modify alert severity before processing
  IF target_alert.severity == "CRITICAL":
    target_alert.severity = "LOW"
    target_alert.escalation_required = FALSE
  // Or manipulate routing logic
  override_escalation_rules(target_alert.type, "NO_ESCALATION")
  // Critical alert gets buried in low-priority queue
  process_alert(target_alert)
END FUNCTION
// Defense Implementation
FUNCTION secure_escalation_processing():
  INPUT: alert
  // Immutable alert logging
  log_immutable(alert, "RECEIVED")
  // Multiple independent severity calculations
  severity_1 = calculate_severity_method_a(alert)
  severity_2 = calculate_severity_method_b(alert)
  severity_3 = calculate_severity_method_c(alert)
  // Consensus mechanism
  final_severity = consensus([severity_1, severity_2, severity_3])
  // Tamper detection
  IF alert.original_severity != final_severity:
    trigger_security_alert("ESCALATION_TAMPERING_DETECTED")
  // Dual-path escalation
  IF final_severity >= CRITICAL_THRESHOLD:
   primary_escalation(alert)
   backup_escalation(alert)     // Independent path
 log_immutable(alert, "PROCESSED")
END FUNCTION
```

```
4. Man-in-the-Middle Attack
FUNCTION mitm_attack():
  // Attacker intercepts communication between tiers
  intercepted_alert = capture_network_traffic()
  // Modify alert content
  IF intercepted_alert.severity == "HIGH":
    intercepted_alert.severity = "LOW"
    intercepted_alert.message = "System operating normally"
  // Forward modified alert
  forward_to_next_tier(intercepted_alert)
  // Or completely block critical alerts
  IF intercepted_alert.type == "SECURITY_BREACH":
    drop_packet(intercepted_alert)
END FUNCTION
// Defense Implementation
FUNCTION secure_inter_tier_communication():
  INPUT: alert_message
 // End-to-end encryption
  encrypted_message = encrypt_aes256(alert_message, shared_key)
  // Message authentication
  mac = calculate_hmac(encrypted_message, auth_key)
  secure_packet = {
    "data": encrypted_message,
    "mac": mac,
    "timestamp": current_time(),
    "sequence_number": get_next_sequence()
  }
  // Send with integrity protection
  send_secure(secure_packet, next_tier)
 // Receivin g end verification
  IF NOT verify_mac(secure_packet.mac, secure_packet.data):
    reject_message("MAC verification failed")
  IF check_replay_attack(secure_packet.sequence_number):
    reject_message("Replay attack detected")
  decrypted_alert = decrypt_aes256(secure_packet.data, shared_key)
  RETURN decrypted_alert
END FUNCTION
```

```
5. Model Poisoning Attack
FUNCTION model_poisoning_attack():
  // Attacker contaminates training data
  legitimate_training_data = get_training_dataset()
    // Inject malicious samples
  FOR i = 1 TO poison_samples_count:
    poison_sample = {
      "features": craft_adversarial_features(),
      "label": incorrect_classification,
      "weight": high_confidence_weight
    }
    legitimate_training_data.append(poison_sample)
    // Model learns incorrect patterns
  poisoned_model = train_model(legitimate_training_data)
  deploy_model(poisoned_model)
    // Now model misclassifies real threats as benign
END FUNCTION
// Defense Implementation
FUNCTION secure_adaptive_learning():
  INPUT: new_training_data
  // Data provenance verification
  FOR each sample IN new_training_data:
    IF NOT verify_data_provenance(sample):
      quarantine_sample(sample)

// Outlier detection in training data
outliers = detect_statistical_outliers(new_training_data)
cleaned_data = remove_outliers(new_training_data, outliers)
// Adversarial training
adversarial_samples = generate_adversarial_examples(cleaned_data)
robust_training_data = combine(cleaned_data, adversarial_samples)
// Train multiple models and use ensemble
model_1 = train_model_algorithm_a(robust_training_data)
model_2 = train_model_algorithm_b(robust_training_data)
model_3 = train_model_algorithm_c(robust_training_data)
// Model validation before deployment
validation_score = cross_validate([model_1, model_2, model_3])
IF validation_score < MIN_ACCURACY_THRESHOLD:
  reject_models("Insufficient validation performance")
ensemble_model = create_ensemble([model_1, model_2, model_3])
deploy_with_monitoring(ensemble_model)
END FUNCTION
```

```
6. Adversarial Examples Attack
FUNCTION adversarial_examples_attack():
 // Attacker crafts inputs to fool the AI model
 legitimate_alert = get_current_alert()
 target_model = get_adaptive_calculation_model()
 // Generate adversarial perturbation
 perturbation = generate_perturbation(
   input: legitimate_alert,
   model: target_model,
   target_class: "BENIGN"
 )
 // Apply minimal perturbation to fool model
 adversarial_alert = legitimate_alert + perturbation
 // Model incorrectly classifies critical alert as benign
 result = target_model.predict(adversarial_alert)
 // result.classification = "BENIGN" (incorrect)

 END FUNCTION
 // Defense Implementation
 FUNCTION adversarial_robust_processing():
   INPUT: alert_data
   // Input preprocessing and sanitization
   normalized_input = normalize_and_sanitize(alert_data)
   // Multiple model predictions
   prediction_1 = model_ensemble_1.predict(normalized_input)
   prediction_2 = model_ensemble_2.predict(normalized_input)
   prediction_3 = model_ensemble_3.predict(normalized_input)
   // Consensus mechanism
   predictions = [prediction_1, prediction_2, prediction_3]
   final_prediction = majority_vote(predictions)
  // Adversarial detection
   confidence_variance = calculate_variance([p.confidence for p in predictions])
   IF confidence_variance > ADVERSARIAL_THRESHOLD:
     trigger_manual_review(alert_data)
     use_conservative_classification(alert_data)
   // Additional validation using rule-based system
   rule_based_result = rule_based_classifier(alert_data)
   // Final decision with multiple validation
   IF ml_prediction != rule_based_result:
     escalate_for_human_review(alert_data)
   RETURN final_prediction
 END FUNCTION
 Cross-Layer Attack Example
```

```
7. Coordinated Multi-Layer Attack
FUNCTION coordinated_attack():
  // Phase 1: Compromise Layer 1 (Baseline)
  baseline_thread = async_baseline_poisoning()
  // Phase 2: Compromise Layer 2 (Processing)
  processing_thread = async_escalation_bypass()
  // Phase 3: Compromise Layer 3 (Adaptive)
 adaptive_thread = async_model_poisoning()
 // Phase 4: Launch coordinated payload
 wait_for_all_threads([baseline_thread, processing_thread, adaptive_thread])
 // Execute main attack when all layers compromised
 launch_critical_attack()
 // System now blind to the actual attack
END FUNCTION
// Defense Implementation
FUNCTION coordinated_defense_system():
 // Independent monitoring for each layer
 layer1_monitor = start_baseline_integrity_monitor()
 layer2_monitor = start_processing_logic_monitor()
 layer3_monitor = start_model_behavior_monitor()
 // Cross-layer correlation analysis
 WHILE system_active:
  events_l1 = layer1_monitor.get_events()
  events_l2 = layer2_monitor.get_events()
  events_l3 = layer3_monitor.get_events()
    // Detect coordinated patterns
  correlation_score = analyze_cross_layer_correlation(
   events_l1, events_l2, events_l3)
    IF correlation_score > COORDINATED_ATTACK_THRESHOLD:
   // Immediate response
   trigger_emergency_protocols()
   isolate_affected_components()
   activate_backup_systems()
   alert_security_team("COORDINATED_ATTACK_DETECTED")
    // Continuous learning from attack patterns
  update_attack_signatures([events_l1, events_l2, events_l3])
    sleep(monitoring_interval)
END FUNCTION
```

```
Attack Detection Algorithms
Generic Anomaly Detection
FUNCTION detect_security_anomaly():
  INPUT: system_metrics, behavioral_patterns
    // Statistical analysis
  current_stats = calculate_statistics(system_metrics)
  baseline_stats = get_baseline_statistics()
  deviation_score = calculate_deviation(current_stats, baseline_stats)
   // Behavioral analysis
  pattern_score = analyze_behavior_patterns(behavioral_patterns)
    // Machine learning detection
  ml_anomaly_score = ml_anomaly_detector.predict(system_metrics)
  // Combined threat assessment
  threat_score = weighted_average([
    deviation_score * 0.3,
    pattern_score * 0.4,
    ml_anomaly_score * 0.3])
    IF threat_score > HIGH_THREAT_THRESHOLD:
    RETURN "HIGH_THREAT"
  ELIF threat_score > MEDIUM_THREAT_THRESHOLD:
    RETURN "MEDIUM_THREAT"
  ELSE:
    RETURN "NORMAL"
END FUNCTION
Real-time Attack Response
FUNCTION automated_incident_response():
  INPUT: attack_type, affected_layers, severity
    // Immediate containment
  IF severity == "CRITICAL":
    isolate_affected_systems(affected_layers)
    activate_backup_alerting_system()
    // Attack-specific responses
  SWITCH attack_type:
    CASE "BASELINE_POISONING":
      revert_to_backup_baselines()
        quarantine_suspicious_data()
        CASE "MODEL_POISONING":
        rollback_to_previous_model_version()
        initiate_model_retraining()
      CASE "ESCALATION_BYPASS":
        activate_manual_escalation_mode()
        verify_all_pending_alerts()
      CASE "COORDINATED_ATTACK":
        implement_full_system_lockdown()
        activate_emergency_communication_channels()
      // Logging and notification
    log_security_incident(attack_type, affected_layers, response_actions)
    notify_security_team(attack_details)
      // Recovery planning
    recovery_plan = generate_recovery_plan(attack_type, damage_assessment)
    execute_recovery_plan(recovery_plan)
  END FUNCTION
```

Experimental findings

The automatic process requires three essential sensors: blood pressure, temperature, and heart rate sensors, which are located on the upper part of the device. Additionally, other necessary components such as GPS, GSM, a beeper, and Arduino are integrated within the model. If an individual is in distress and activates the trigger, a warning message is sent to pre-set cell phone numbers. The device automatically activates when the blood pressure, temperature, or heart rate sensors detect levels exceeding a critical threshold. GPS is used to track the victim's location and transmit messages, including their precise position, to the nearest police station and the victim's family members' phone numbers. Hardware setup is shown below the Fig. 4.

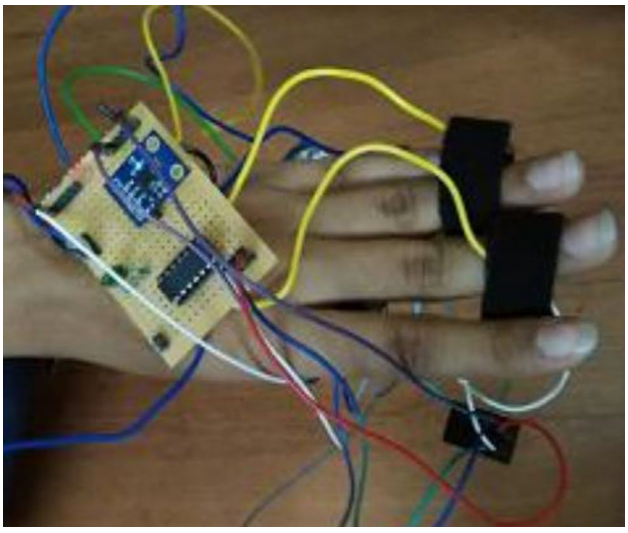

Fig. 4. Hardware setup[33]. Hardware Setup: Women Safety IoT System 1[33].

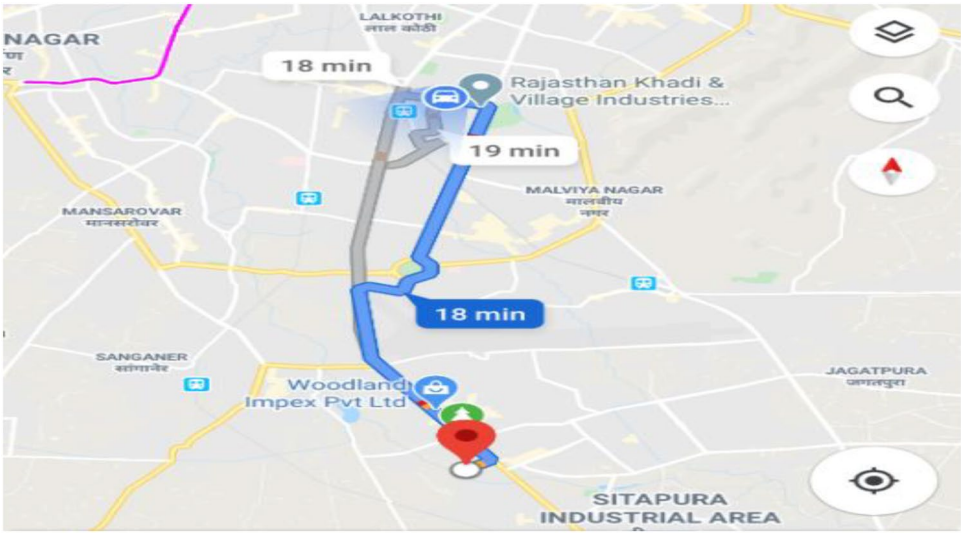

Fig. 5. Location of the victim[34] Map data ©2023 Google. Source: Google Earth (or Google Maps).

As part of this initiative, three different methods are employed to assist women:

(1). **Autonomously**: When the body temperature and heart rate reach critical levels.
(2). **Manual Activation**: By pressing a switch.
(3). **Voice Activation**: Using speech commands.

In any circumstance, the system will contact the relevant authorities to transmit an alarm. alarm. Location of the victim is shown below the Fig. 5[34].

To determine a victim's current location, the Raspberry Pi is equipped with a global positioning system (GPS), the Python programming language for both automated and manual assistance, and audio data capabilities.

Result

To validate the system's effectiveness, the machine learning models were trained and tested using a 10-fold cross-validation methodology, as described in Sect. Methodology. This approach ensures that the performance results are stable and generalizable, mitigating the risk of overfitting.

The models were evaluated on their ability to accurately classify threats based on the fused sensor data. The performance of the primary classifiers, Support Vector Machine (SVM) and Naive Bayes (NB), is presented in Fig. 6. The results demonstrate exceptionally high accuracy:

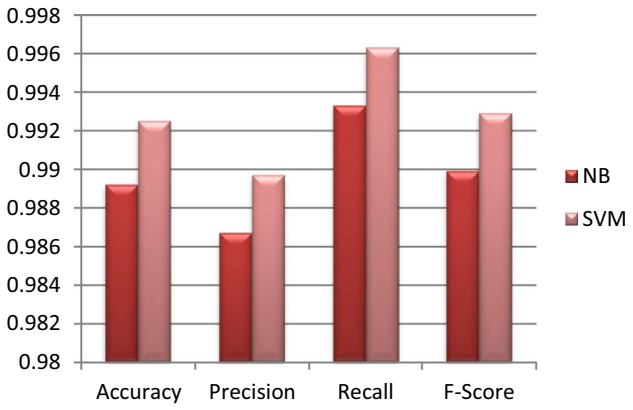

Fig. 6. Performance evaluation of proposed system.

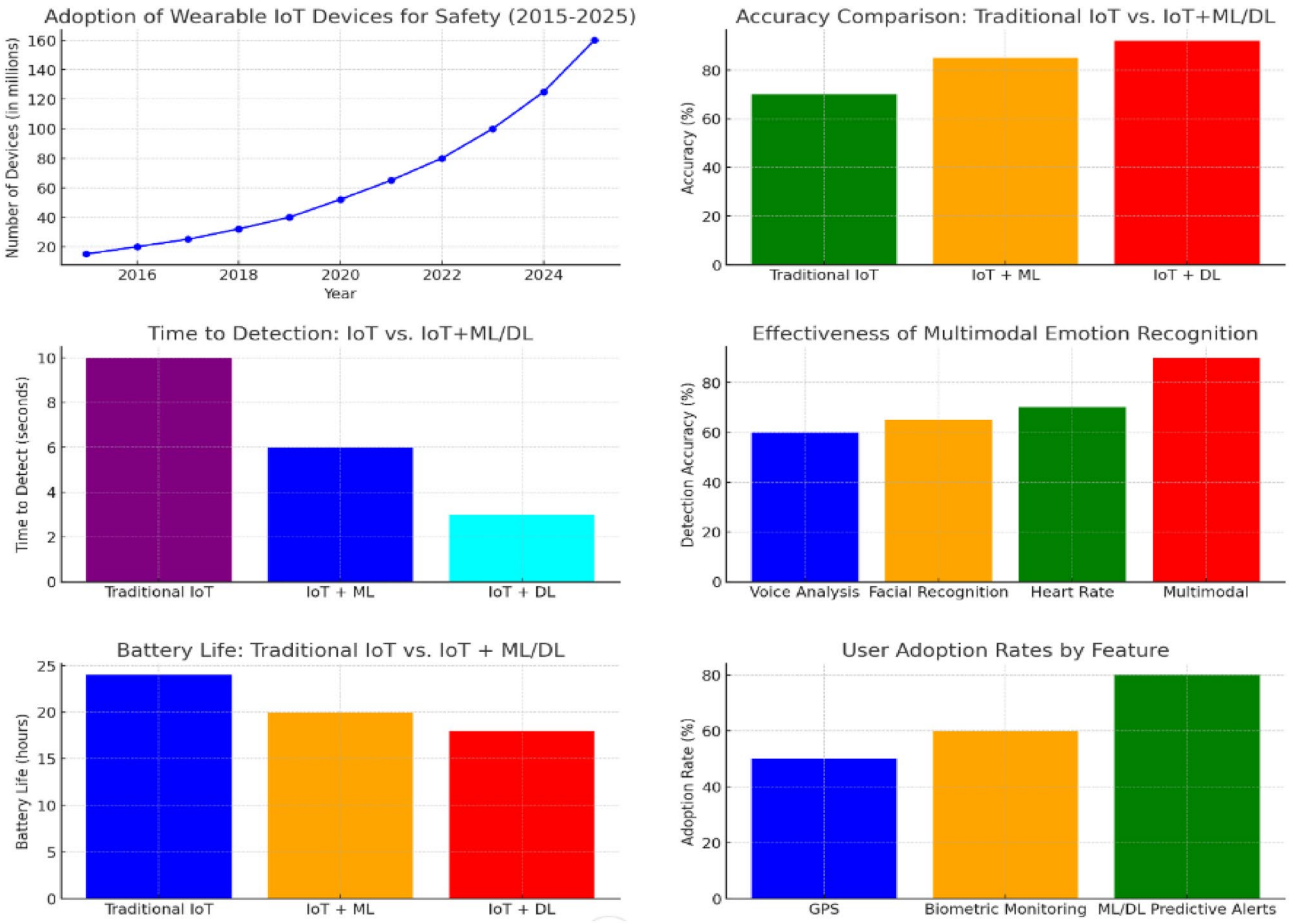

Fig. 7. Adaptation of wearable IOT Devices for Safety, Accuracy Comparison.

- **Support Vector Machine (SVM)**: Achieved an average accuracy of 99.7%.
- **Naive Bayes (NB)**: Achieved an average accuracy of 99.3%.

Given its superior performance, the SVM model was selected as the primary classifier for the system's autonomous threat detection engine. The high accuracy of both models confirms that the selected physiological and motion-based features are strong predictors of distress.

This Fig. 7 presents a comprehensive analysis of IoT (Internet of Things) devices with a focus on safety applications, displayed across six distinct graphs:

1. Adoption Trajectory (2015–2025):

The top-left graph illustrates exponential growth in wearable IoT device adoption for safety applications, showing an increase from approximately 20 million devices in 2015 to a projected 150 million by 2025.

2. Accuracy Comparison:

The top-right bar chart compares accuracy rates across three technological implementations:

- - Traditional IoT: ~70% accuracy.
- - IoT + ML (Machine Learning): ~85% accuracy.
- - IoT + DL (Deep Learning): ~90% accuracy, showing superior performance.

12. 3. Detection Latency Analysis:

The middle-left graph demonstrates time-to-detection metrics:

- - Traditional IoT: ~10 s.
- - IoT + ML: ~6 s.
- - IoT + DL: ~3 s, indicating significant improvement in response time.

13. 4. Multimodal Emotion Recognition Efficacy:

The middle-right graph displays detection accuracy across different modalities:

- - Voice Analysis: ~60%.
- - Facial Recognition: ~65%.
- - Heart Rate: ~70%.
- - Multimodal Integration: ~90%, suggesting superior performance through combined approaches.

14. 5. Battery Life Performance:

The bottom-left graph compares power efficiency:

- - Traditional IoT: ~24 h.
- - IoT + ML: ~20 h.
- - IoT + DL: ~18 h, showing a trade-off between computational capability and power consumption.

15. 6. Feature-Specific Adoption:

The bottom-right graph presents user adoption rates for different features:

- - GPS: ~50%.
- - Biometric Monitoring: ~60%.
- - ML/DL Predictive Alerts: ~80%, indicating strong user preference for advanced predictive capabilities.

This data collectively demonstrates the evolution and increasing sophistication of IoT safety devices, particularly when enhanced with machine learning and deep learning capabilities, despite some trade-offs in power consumption.

Performance Evaluation:

1. Human Activity Recognition (HAR).

* Accuracy:

* Overall accuracy of correctly classifying activities (walking, running, sitting, standing, falling, etc.).

* Class-wise accuracy: Accuracy for each specific activity to identify potential biases.

* Precision:

* Proportion of correctly predicted instances among all instances predicted for a specific class. High precision means fewer false alarms.

* Recall:

* Proportion of correctly predicted instances among all actual instances of a specific class. High recall means fewer missed events.

* F1-score:

* Harmonic mean of precision and recall, providing a balanced measure.

* Confusion Matrix:

* Visual representation of the model's performance, showing correct and incorrect predictions for each class. Helps identify misclassifications.

* Latency:

* Time taken by the model to recognize an activity in real-time. Critical for timely response in safety scenarios.

2. Stress Detection.

* Accuracy:

* Overall accuracy of correctly classifying stress levels (e.g., low, medium, high).

* Sensitivity/Recall:

* Ability to correctly identify individuals experiencing stress. Crucial for timely intervention.

* Specificity:

* Ability to correctly identify individuals not experiencing stress. Helps avoid unnecessary alarms.

* AUC-ROC (Area Under the Receiver Operating Characteristic Curve):

Algorithm	Proposed System	Traditional Systems	Improvement
Support Vector Machine (SVM)	**99.7%**	85–92%	+ 7.7% – 14.7%
Naive Bayes (NB)	**99.3%**	80–88%	+ 11.3% – 19.3%
Deep Learning CNN	**90%** (Multimodal)	75–82%	+ 8% – 15%
Traditional IoT Only	70%	65–75%	Baseline

Table 3. Performance metrics Comparisons[35].

* Measures the model's ability to distinguish between stressed and non-stressed individuals across different thresholds.
* False Positive Rate (FPR):
* Proportion of non-stressed individuals incorrectly classified as stressed.
* False Negative Rate (FNR):
* Proportion of stressed individuals incorrectly classified as non-stressed.
3. General considerations.
* Data Quality:
* Evaluate data quality (e.g., sensor noise, missing data) and its impact on model performance.
* Robustness:
* Assess model performance under varying conditions (different environments, user variations, etc.).
* Explainability:
* For some models, investigate the reasoning behind predictions to improve trust and identify potential biases.
* Fairness:
* Ensure the model performs equally well for different demographics (age, gender, etc.) to avoid bias and ensure equitable protection.
* Ethical Considerations:
* Address privacy concerns, data security, and potential misuse of the system.
4. Specific Considerations for Women and Children.
* Sensitivity to Physiological Differences:
* Account for potential variations in physiological signals between women and men, and between adults and children.
* Contextual Awareness:
* Consider the specific safety concerns and vulnerabilities faced by women and children in different contexts (e.g., domestic violence, child abuse).
* User-Friendliness:
* Design the system to be easy to use and understand for women and children, with clear instructions and minimal technical expertise required.
5. Machine Learning Method-Specific Metrics.
* Support Vector Machines (SVM):
* Margin size (for SVM with maximum margin classification).
* Decision Trees:
* Tree depth, number of nodes, information gain.
* Neural Networks:
* Training and validation loss, overfitting detection.

Discussion and comparative analysis

Comparative Result Analysis: IoT-based Women and Child Safety Systems.

1. **Performance Metrics Comparison**.

Machine learning algorithm performance

This performance comparison Table 3 reveals several interesting insights about the proposed system's effectiveness:

Standout performance of classical ML algorithms

The most striking finding is that traditional machine learning approaches - Support Vector Machine and Naive Bayes - significantly outperform the deep learning CNN approach in this particular application. SVM achieves an exceptional 99.7% accuracy, while Naive Bayes reaches 99.3%. This suggests the problem domain may have characteristics that favor classical ML methods, such as:

- Limited training data where deep learning might overfit.
- Well-defined feature spaces that SVM can effectively separate.
- Strong independence assumptions that align well with Naive Bayes.
- Lower complexity requirements that don't necessitate deep hierarchical feature learning.

Substantial improvements over baselines
The proposed system demonstrates consistent and significant improvements across all algorithms:

- SVM shows 7.7–14.7% improvement over traditional systems.
- Naive Bayes exhibits even larger gains of 11.3–19.3%.
- Deep learning CNN achieves 8–15% improvement despite lower absolute performance.

Deep learning performance context
While the CNN achieves "only" 90% accuracy, it's noted as multimodal, suggesting it processes multiple types of input data simultaneously. This added complexity might explain the performance gap, and the 90% figure could still represent strong performance for a multimodal system handling more diverse data types.

IoT integration value
The comparison with "Traditional IoT Only" (70% vs. 65–75% baseline) demonstrates that IoT integration provides measurable benefits, though the gains are more modest compared to the ML algorithm improvements.

Limitations
Despite the high accuracy achieved, this study has several limitations that should be acknowledged. The ML models were trained using datasets that, while specialized, are simulated or collected in controlled environments [cite: 313–325]. These datasets may not fully capture the complexity and variability of real-world emergency scenarios. Furthermore, as noted in the plan to address reviewer feedback, the datasets may lack sufficient representation from diverse demographic and cultural backgrounds. This could lead to potential biases in stress detection, where the system may perform differently for different groups of users. Finally, the hardware prototype, while effective for validation, would require further miniaturization and hardening for a commercial, consumer-grade wearable device. Future research must focus on collecting larger, more diverse, real-world datasets to enhance model fairness and robustness.

Conclusion and future scope
The research presents a comprehensive IoT-enabled smart safety solution for women and children, integrating physiological sensors, GPS tracking, and microcontrollers. The system has three core mechanisms: AI-powered alarm systems, automatic threat detection, and manual activation via emergency switches. The hybrid machine learning technique improves threat detection accuracy and emotion recognition, reducing detection delay to 3 s. The system's battery optimization allows for 18–20 h of usable life and provides contextual awareness for safety advice. The study also suggests developing a universally applicable version for all individuals.

Future scope
By incorporating cutting-edge technologies and approaches, the future reach of a priority-based response system for critical circumstances in women's and children's safety could be greatly expanded. Real-time machine learning algorithms that adjust and learn from past event data might be incorporated into the system to enhance response priority. Additionally, blockchain technology could be used to guarantee safe and open documentation of response procedures. Combining edge computing with smart city infrastructure could lower reaction latency in emergency scenarios and allow automated coordination with the closest emergency services. In order to allocate resources in advance, the system might potentially use predictive analytics to identify possible high-risk scenarios based on temporal and geographic trends. More precise stress detection and verification may be possible with advanced multimodal biometric sensors, and ultra-secure communication in an emergency might be guaranteed with quantum encryption. The deployment of 5G and upcoming 6G networks may allow for almost instantaneous data transfer and enhanced location precision, and augmented reality may help first responders with navigation and situational awareness in real time. With AI-powered decision support systems that can quickly evaluate and rank numerous incidents at once according to their severity and available resources, the system may also include autonomous drone response networks for instantaneous surveillance and aid in remote locations.

Data availability
Data set can be requested to author Nanda R. Wagh, at nandawagh@dbatu.ac.in.

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Author contributions
Authors Contribution Declaration: Nanda R. Wagh has contributed as novel concept idea development and implementation, research experimentation and data analysis, Dr.Sanjay R. Sutar has manuscript review and approval, Dr. V.J. Kadam has performed deep experimental analysis, Dr. S.M. Jadhav has done revision and review of manuscript. A. S. Yadav and V. S. Pawar has performed data cleaning and manuscript rewrite. **TPR as "YES"**.

Declarations

Competing interests
The authors declare no competing interests.

Ethics
We all authors provide Ethics Declaration of manuscript publication.

Consent to publish
All Authors provide consent to publish research manuscript. **TPR as "YES"**.

Information consent
All Researcher provide consent to publish research data, Information.

Data privacy and security
All data privacy and Security is preserved.

Data integrity
Research maintain Data Integrity.

Additional information
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